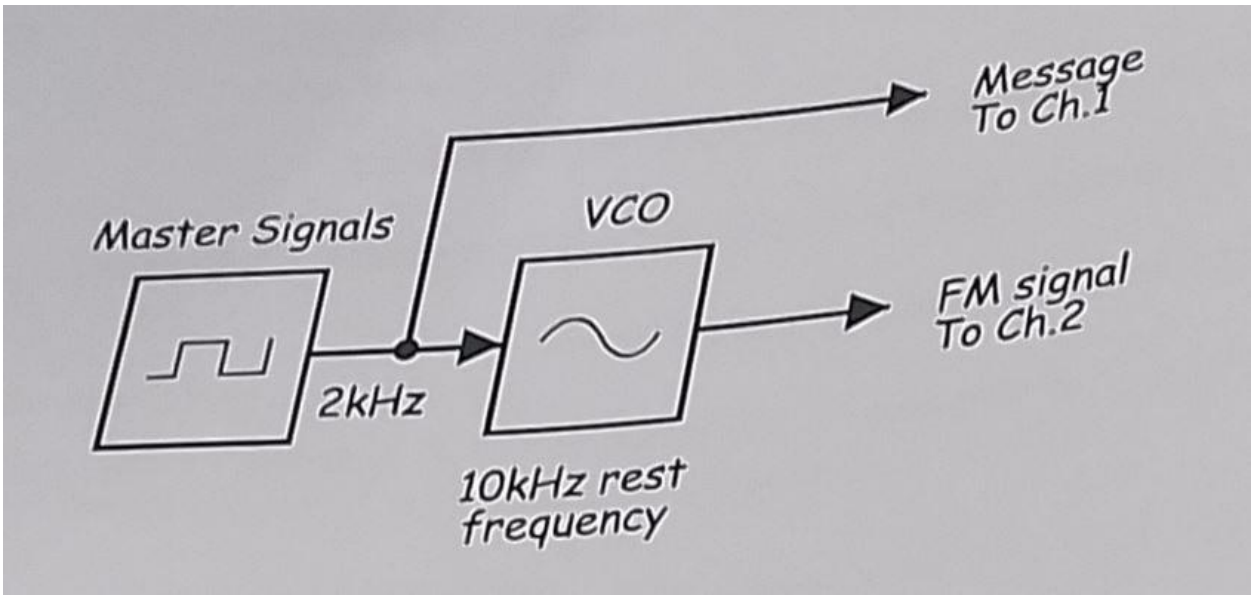
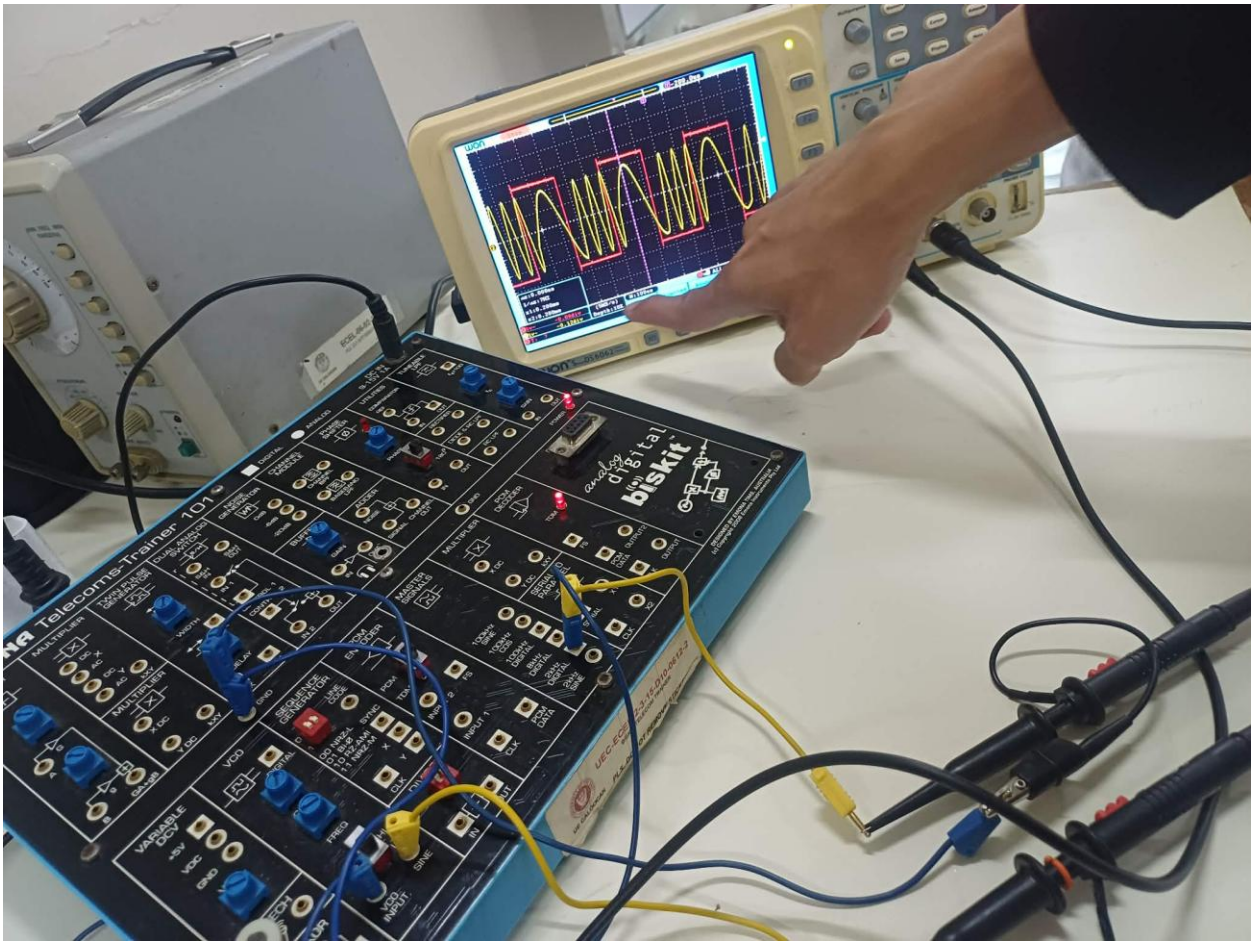
Circuit Block Diagram:

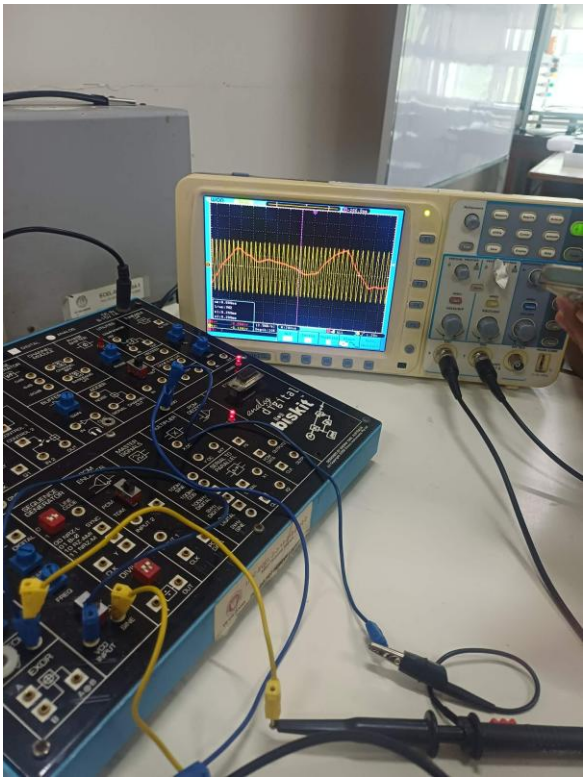
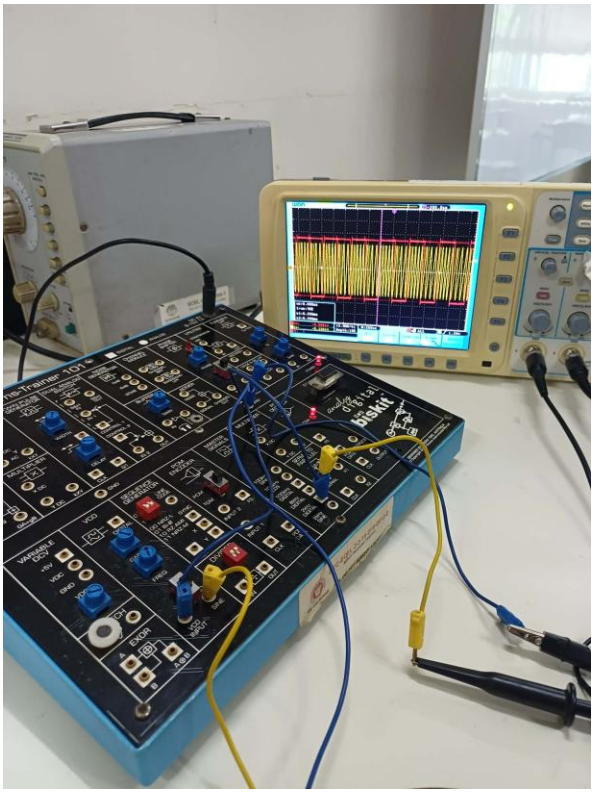


Circuit Set Up & Result:

Part A:



Part B:



Part C: